

Spring 2026 OOP HW 01

Part 1

Research the latest uses of C++ — What is C++ currently used for, and why is it still preferred over other programming languages? Write a short summary (3–4 lines) or present it in a small table.

C++ is known for its high performance and direct control over hardware and memory, making it ideal for performance-critical applications even today. It is widely used in systems software and embedded system development, where low-level access is essential. C++ is also a dominant language for game engines and high-frequency trading systems, where low latency and fine-grained optimization are crucial. Overall, its minimal runtime overhead and mature ecosystem keep C++ a preferred choice for building efficient and reliable software.

Research the best IDE or AI tool for writing C++ code — Identify and compare the top 3 IDEs or AI tools you find, focusing on the 3–4 most important factors you consider relevant. Present your comparison in a clear table.

Tool	C++ Support and Code Intelligence	Difficulty to Use and Setup	Extensibility and Integration	Cost, Accessibility, and Availability
Visual Studio	Excellent native C++ support, strong tools like IntelliSense for coding and debugging	Mostly ready out of the box, and has moderate and rewarding learning curve; might be a bit bloated or heavy	Rich and powerful built-in tools and extensions	Free Community edition; paid Professional/Enterprise versions; Windows only
VS Code	NO native C++ support but strong C++ support through extensions	Really easy to install, and lightweight	Very large extension marketplace, offering outstanding flexibility	Free; available on Windows, Linux, and macOS
GitHub Copilot	AI-based code completion and suggestions for C++ and many other languages	Very easy to use as a plugin, for in-line code completion, dialogue-driven code generation, and debugging	Already integrated with IDEs like VS and VS Code	Subscription-based; free for students and limited trial available.

Select one of these tools to complete Part 2 of the assignment — Important: If an AI tool was used to create the program, ensure you fully understand the generated code, as TAs may request a follow-up explanation or verification. Failure to explain the code when asked will result in a score of zero.

HW01 is completed by VS Code and with some assistant from integrated github copilot pro.